

Monthly Temperature Derivation

Generate a monthly table of average temperature per station from the daily data, for use as a new fact table in Power BI.

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: df = pd.read_csv("../Capstone/clean data/dim_temp_clean.csv")
```

C:\Users\biauser\AppData\Local\Temp\ipykernel_4680\3186851481.py:1: DtypeWarning: Columns (0) have mixed types. Specify dtype option on import or set low_memory=False.
 df = pd.read_csv("../Capstone/clean data/dim_temp_clean.csv")

```
In [3]: df.head()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 833189 entries, 0 to 833188
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Climate ID            833189 non-null  object
1   Date/Time             833189 non-null  object
2   Max Temp (°C)         828481 non-null  float64
3   Min Temp (°C)         831117 non-null  float64
4   Mean Temp (°C)        833189 non-null  float64
5   Heat Deg Days (°C)    826411 non-null  float64
6   Cool Deg Days (°C)    826411 non-null  float64
dtypes: float64(5), object(2)
memory usage: 44.5+ MB
```

```
In [4]: # Fix key data types early
df["Climate ID"] = df["Climate ID"].astype(str)
```

```
In [6]: cols = [
    "Climate ID",
    "Date/Time",
    "Mean Temp (°C)"
]

df_temp = df[cols].copy()
```

```
In [7]: df_temp["Date/Time"] = pd.to_datetime(df_temp["Date/Time"], errors="coerce")

df_temp["Year"] = df_temp["Date/Time"].dt.year
df_temp["Month"] = df_temp["Date/Time"].dt.month

# Forma recomendada para BI
df_temp["YearMonth"] = df_temp["Date/Time"].dt.to_period("M").astype(str)
```

```
In [8]: df_temp = df_temp.dropna(
        subset=["Mean Temp (°C)", "Year", "Month"]
    )
```

Monthly aggregation (notebook core)

 Clear methodological definition:

Monthly temperature = average of daily mean temperatures

```
In [9]: monthly_temp = (
        df_temp
        .groupby(
            ["Climate ID", "Year", "Month", "YearMonth"],
            as_index=False
        )
        ["Mean Temp (°C)"]
        .mean()
    )
```

```
In [10]: monthly_temp = monthly_temp.rename(
        columns={"Mean Temp (°C)": "MeanMonthlyTemp_C"}
    )
```

```
In [11]: monthly_temp.head(10)
```

```
Out[11]:
```

	Climate ID	Year	Month	YearMonth	MeanMonthlyTemp_C
0	8100300	1995	1	1995-01	-9.061290
1	8100300	1995	2	1995-02	-13.600000
2	8100300	1995	3	1995-03	-3.287097
3	8100300	1995	4	1995-04	0.960000
4	8100300	1995	5	1995-05	10.483871
5	8100300	1995	6	1995-06	17.320000
6	8100300	1995	7	1995-07	20.477419
7	8100300	1995	8	1995-08	18.396774
8	8100300	1995	9	1995-09	12.820000
9	8100300	1995	10	1995-10	10.119355

```
In [12]: monthly_temp.describe()
monthly_temp["YearMonth"].nunique()
monthly_temp["Climate ID"].nunique()
```

```
Out[12]: 136
```

```
In [13]: monthly_temp[
          monthly_temp["Climate ID"] == "8204020"
          ].sort_values("YearMonth")
```

Out[13]:

	Climate ID	Year	Month	YearMonth	MeanMonthlyTemp_C
11285	8204020	2021	9	2021-09	15.330000
11286	8204020	2021	10	2021-10	10.487097
11287	8204020	2021	11	2021-11	3.696667
11288	8204020	2021	12	2021-12	-1.306452
11289	8204020	2022	1	2022-01	-6.740000
11290	8204020	2022	2	2022-02	-4.500000
11291	8204020	2022	3	2022-03	0.020000
11292	8204020	2022	4	2022-04	4.936667
11293	8204020	2022	5	2022-05	11.358065
11294	8204020	2022	6	2022-06	15.143333
11295	8204020	2022	7	2022-07	19.816129
11296	8204020	2022	8	2022-08	20.253333
11297	8204020	2022	9	2022-09	14.800000
11298	8204020	2022	10	2022-10	10.974194
11299	8204020	2022	11	2022-11	4.826667
11300	8204020	2022	12	2022-12	0.096774
11301	8204020	2023	1	2023-01	-1.460000
11302	8204020	2023	2	2023-02	-6.192857
11303	8204020	2023	3	2023-03	-1.009677
11304	8204020	2023	4	2023-04	5.573333
11305	8204020	2023	5	2023-05	9.364516
11306	8204020	2023	6	2023-06	14.913333
11307	8204020	2023	7	2023-07	21.322581
11308	8204020	2023	8	2023-08	17.600000
11309	8204020	2023	9	2023-09	16.200000
11310	8204020	2023	10	2023-10	11.038710
11311	8204020	2023	11	2023-11	1.133333
11312	8204020	2023	12	2023-12	-0.051613
11313	8204020	2024	1	2024-01	-4.719355
11314	8204020	2024	2	2024-02	-3.706897

	Climate ID	Year	Month	YearMonth	MeanMonthlyTemp_C
11315	8204020	2024	3	2024-03	1.541379
11316	8204020	2024	4	2024-04	4.396552
11317	8204020	2024	5	2024-05	11.725806
11318	8204020	2024	6	2024-06	16.493333
11319	8204020	2024	7	2024-07	20.660000
11320	8204020	2024	8	2024-08	19.131034
11321	8204020	2024	9	2024-09	15.211111
11322	8204020	2024	10	2024-10	9.619355
11323	8204020	2024	11	2024-11	4.073333
11324	8204020	2024	12	2024-12	-2.306897
11325	8204020	2025	1	2025-01	-6.700000
11326	8204020	2025	2	2025-02	-6.725000
11327	8204020	2025	3	2025-03	1.061290
11328	8204020	2025	4	2025-04	4.307692
11329	8204020	2025	5	2025-05	11.452000
11330	8204020	2025	6	2025-06	16.024138
11331	8204020	2025	7	2025-07	20.040000
11332	8204020	2025	8	2025-08	18.306452
11333	8204020	2025	9	2025-09	15.400000
11334	8204020	2025	10	2025-10	10.476667
11335	8204020	2025	11	2025-11	2.913333
11336	8204020	2025	12	2025-12	-4.422581
11337	8204020	2026	1	2026-01	-5.929032
11338	8204020	2026	2	2026-02	-4.610714
11339	8204020	2026	3	2026-03	-0.819355
11340	8204020	2026	4	2026-04	3.819231

```
In [14]: output_path = "../Capstone/clean data/monthly_temperature_by_station.csv"

monthly_temp.to_csv(
    output_path,
    index=False
)
```

```
print(f"Monthly temperature file saved at: {output_path}")
```

Monthly temperature file saved at: ../Capstone/clean data/monthly_temperature_by_station.csv

```
In [15]: monthly_temp.info()
monthly_temp.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 28609 entries, 0 to 28608
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Climate ID            28609 non-null  object
1   Year                  28609 non-null  int32
2   Month                 28609 non-null  int32
3   YearMonth              28609 non-null  object
4   MeanMonthlyTemp_C     28609 non-null  float64
dtypes: float64(1), int32(2), object(2)
memory usage: 894.2+ KB
```

```
Out[15]:
```

	Climate ID	Year	Month	YearMonth	MeanMonthlyTemp_C
0	8100300	1995	1	1995-01	-9.061290
1	8100300	1995	2	1995-02	-13.600000
2	8100300	1995	3	1995-03	-3.287097
3	8100300	1995	4	1995-04	0.960000
4	8100300	1995	5	1995-05	10.483871

```
In [16]: monthly_temp[["Climate ID", "YearMonth"]].drop_duplicates().shape
```

```
Out[16]: (28609, 2)
```